

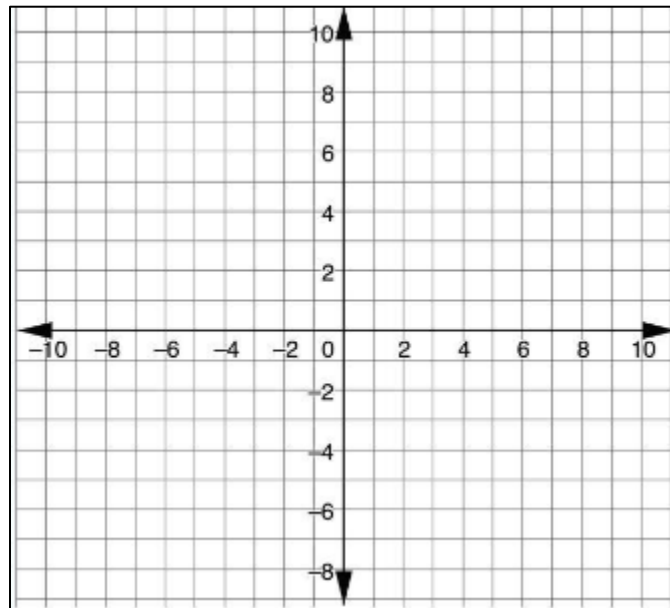
Geometry
Unit 13
Lesson 1 Practice

Name _____
Date _____
Period _____

1. Graph the equations on the coordinate plane:

$$y = -\frac{1}{2}x + 7$$

$$y = 2x - 3$$



2. The equations $y = -\frac{1}{2}x + 7$ and $y = 2x - 3$ resulted in

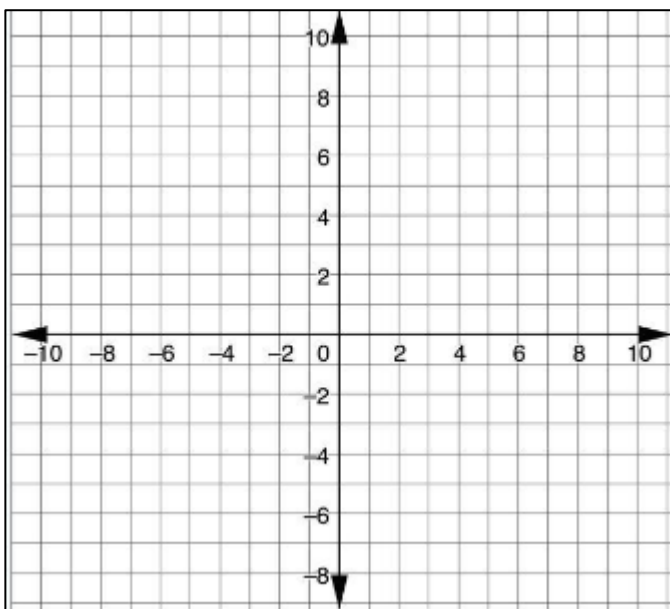
- ☐ parallel
- ☐ perpendicular
- ☐ vertical

lines.

3. The slopes of the equations are

- ☐ equal.
- ☐ undefined.
- ☐ opposite reciprocals.

4. Rewrite the equation $7y - x = -28$ in slope intercept form and graph the equation using slope-intercept form.

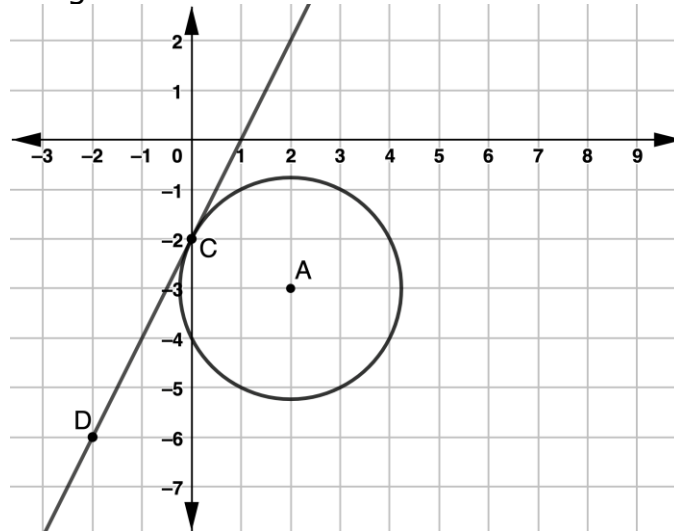


5. Determine if $y = -7x + 4$ is parallel, perpendicular, or neither to the graph of the equation $7y - x = -28$.
6. Graph the equation $y = -7x + 4$ on the graph to verify your answer to question 5.
7. Which equation represents a line which is perpendicular to the line $y = \frac{3}{8}x - 1$?
- A. $3x + y = -40$
 - B. $3x - 8y = -56$
 - C. $3y - 8x = 3$
 - D. $8x + 3y = 9$
8. Which equation represents a line which is parallel to the line $y = -\frac{2}{3}x + 1$?
- A. $2x - 3y = 21$
 - B. $2y - 3x = 2$
 - C. $3x - 2y = -2$
 - D. $2x + 3y = -21$

Geometry
Unit 13
Lesson 2 Practice

Name _____
Date _____
Period _____

Use the following information for questions 1-4. Circle A is shown, where point C is on the circle and line CD is tangent to circle A .



1. Use a straightedge to draw a line that goes through points A and C .
2. Write the equation of line CD .
3. Write the equation of line AC .
4. Line CD and line AC intersect at point $(0, -2)$. Use the equations of line CD and AC to confirm this point.

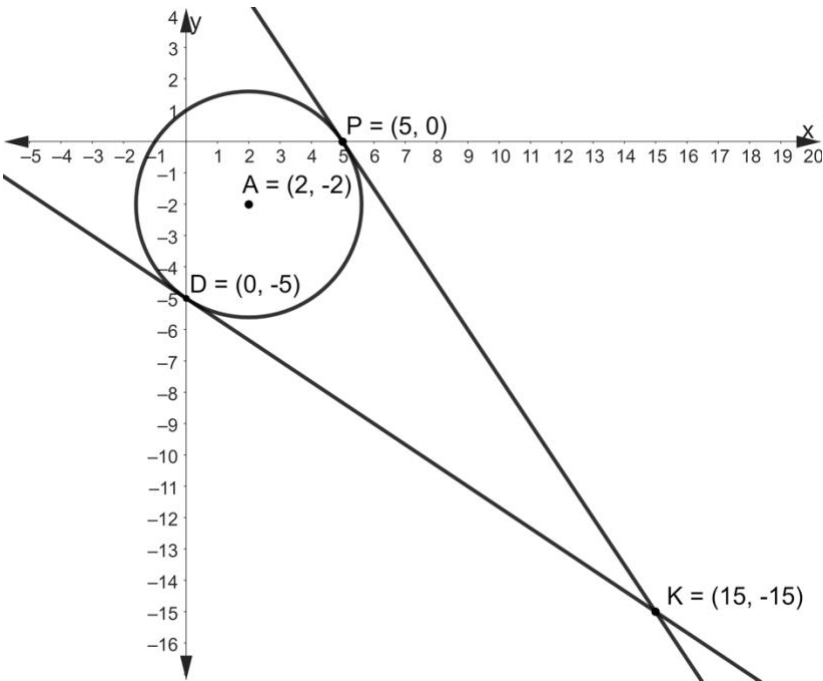
Use the following information for questions 5-8. Circle P has a center at $(4, -3)$ and a tangent line with the equation $y = 4x - 2$.

- 5. Write the equation of the line perpendicular to the tangent line.
- 6. Find the point where the tangent line touches circle P . Call the point H .
- 7. What is the radius of circle P ?
- 8. Write the equation of circle P .

Use the following information for questions 9-10. Circle A is shown with tangent lines PK and DK .

- 9. Complete the table.

Line Segment	Length
$\overline{PK}$	
$\overline{DK}$	
$\overline{AD}$	



- 10. Write the equation of circle A .

Geometry
Unit 13
Lesson 3 Practice

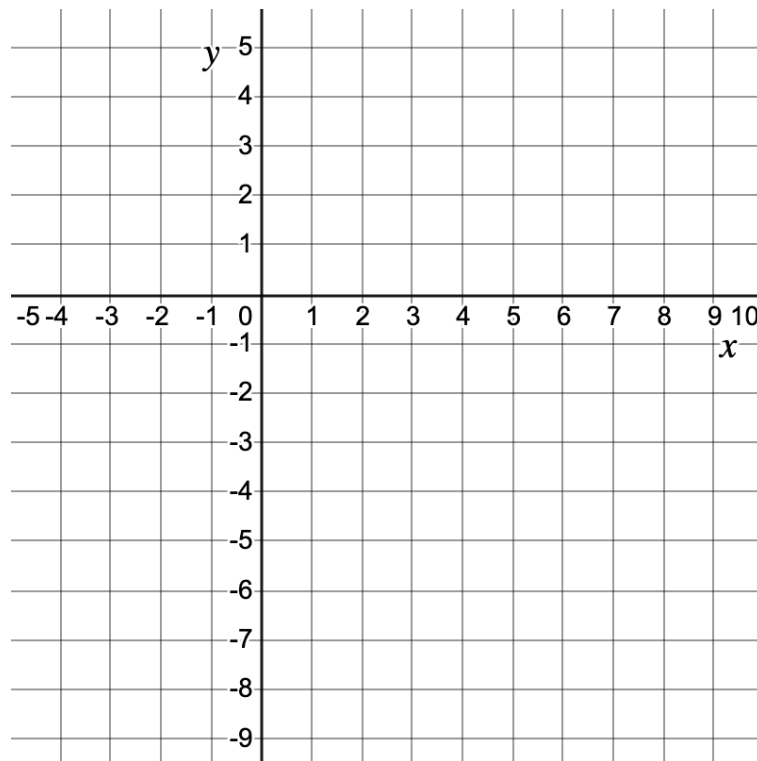
Name _____
Date _____
Period _____

The equation of circle C is $(x - 6)^2 + (y + 3)^2 = 9$.

1. Complete the table by substituting the given coordinates for x and y in the equation of circle C . Then, determine the value of the left side of the equation to complete the simplified equation in the last column.

Coordinate	Substitution	Simplified Equation
$R(6,0)$		_____ = 9
$G(3,-3)$		_____ = 9
$T(6,-6)$		_____ = 9

2. Graph circle C , $(x - 6)^2 + (y + 3)^2 = 9$.



3. Plot $R(6,0)$, $G(3,-3)$, $T(6,-6)$ on the circle.
4. Determine the length of the diameter on circle C .

Circle D has a radius of 2 with points $L(-3,7)$, $F(-1,5)$, and $T(1,7)$ on the circle.

5. Determine which two points are the endpoints of a diameter of circle D .
6. Find the center of circle D .
7. Write the equation for circle D .
8. The equation of circle W is $(x + 7)^2 + y^2 = 4$. Determine whether the following points lie inside, outside, or on circle W .

Point	Work	Location
$C(-7,2)$		☐ Inside circle W ☐ Outside circle W ☐ On circle W
$N(-6,1)$		☐ Inside circle W ☐ Outside circle W ☐ On circle W
$T(-7,5)$		☐ Inside circle W ☐ Outside circle W ☐ On circle W

9. The equation of circle R is $(x - 4)^2 + (y + 2)^2 = 13$.

Select all points that lie on circle R .

- ☐ $(6,1)$
 - ☐ $(4,-2)$
 - ☐ $(1,0)$
 - ☐ $(0,0)$
 - ☐ $(-4,-2)$
10. Circle M and circle T intersect at point H .

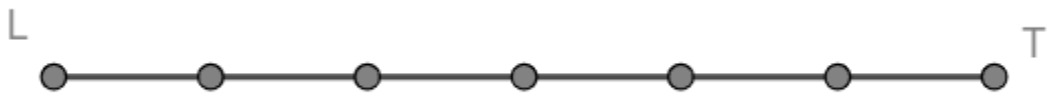
Equation of circle M is $x^2 + (y - 2)^2 = 10$

Equation of circle T is $(x + 4)^2 + (y - 2)^2 = 2$

Which coordinate pair is the location of point H .

- ☐ $(-5,2)$
- ☐ $(0,0)$
- ☐ $(-3,0)$
- ☐ $(-3,1)$

Use the following information for questions 1-4. Directed line segment $\overline{LT}$ is shown.



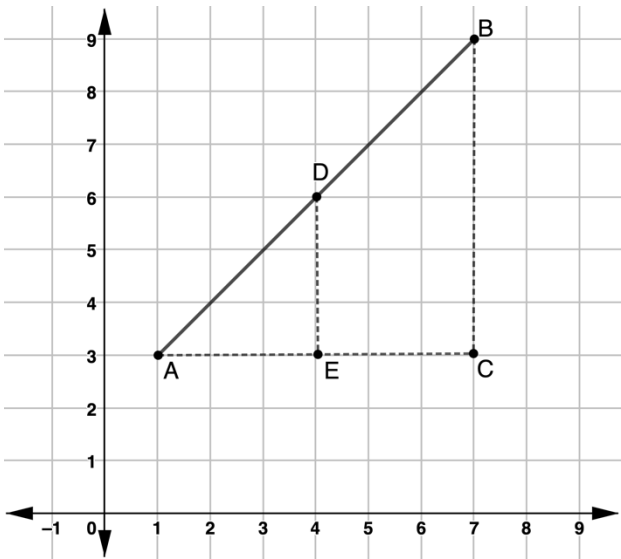
- 1. How many equal parts is $\overline{LT}$ partitioned into?
- 2. Make a mark where $\overline{LT}$ is partitioned into a 1:3 ratio. Call this point K .
- 3. Complete the statement.
Point K is _____ of the distance from L to T .
- 4. Complete the equation that can be used to find the distance of $\overline{LT}$.

$LK = \underline{\hspace{1cm}} LT$

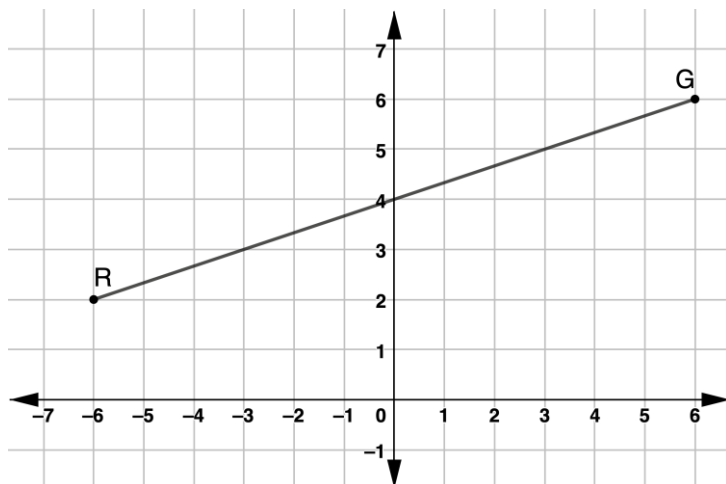
- 5. Point D partitions directed line segment $\overline{AB}$ in a 3:3 ratio. Points $A(1,3)$, $E(4,3)$, $D(4,6)$, $B(7,9)$, and $C(7,3)$ are shown on the coordinate grid.

Use the distances to write the numeric ratio for each segment ratio. Then, choose the type of ratio the segment ratio represent.

Segment Ratio	Numeric Ratio	Type of Ratio
$AE:EC$		☐ part: part ☐ part: whole
$AE:AC$		☐ part: part ☐ part: whole
$DE:BC$		☐ part: part ☐ part: whole
$AD:DB$		☐ part: part ☐ part: whole



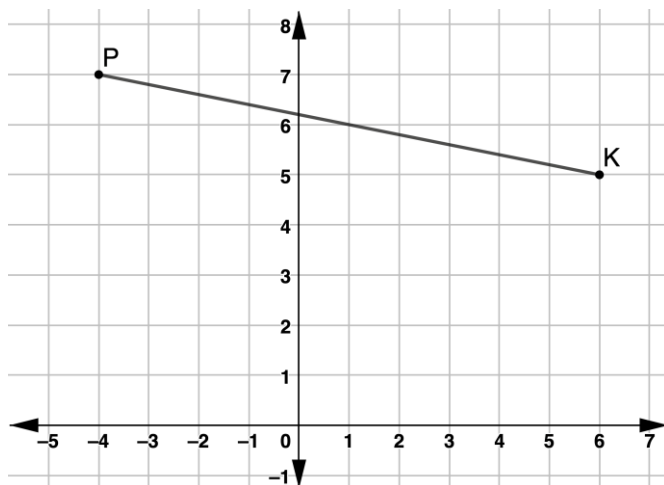
Use the following information for questions 6-8. Directed line segment RG is shown with endpoints at point $R(-6,2)$ and point $G(6,6)$.



Directed line segment RG is partitioned $\frac{3}{4}$ of the distance from R to G at point T .

6. What is the horizontal distance of $\overline{RG}$?
7. What is the vertical distance of $\overline{RG}$?
8. What are the coordinates of point T ?

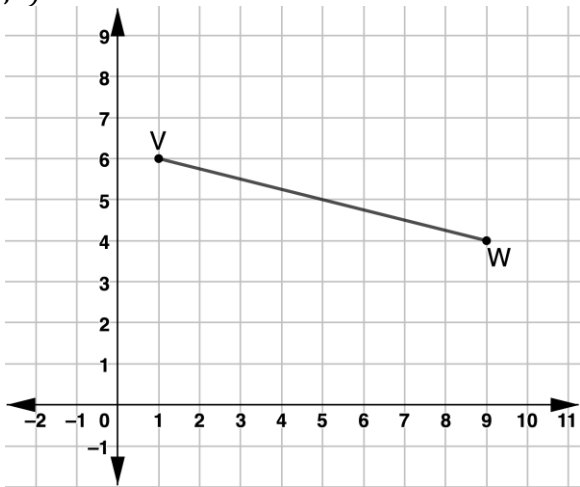
Use the following information for questions 9-10. Directed line segment PK has endpoints at $P(-4,7)$ and $K(6,5)$.



Directed line segment PK is partitioned $\frac{2}{5}$ of the distance from P to K at point R .

9. What is the horizontal and vertical distance of $\overline{PK}$?
10. What are the coordinates of point R ?

Use the following information for questions 1-4. Point V has coordinates $(1,6)$ and point W has coordinates $(9,4)$.



1. Complete the table to calculate $\frac{1}{2}V + \frac{1}{2}W$.

	<i>x</i> -value	<i>y</i> -value
$V(1,6)$	$\frac{1}{2}(\text{---})$	$\frac{1}{2}(\text{---})$
$W(9,4)$	$\frac{1}{2}(\text{---})$	$\frac{1}{2}(\text{---})$
Total		

2. What is the midpoint of line segment VW ?
3. Complete the expression that can be used to find the point that partitions the line segment in a 1:1 ratio.

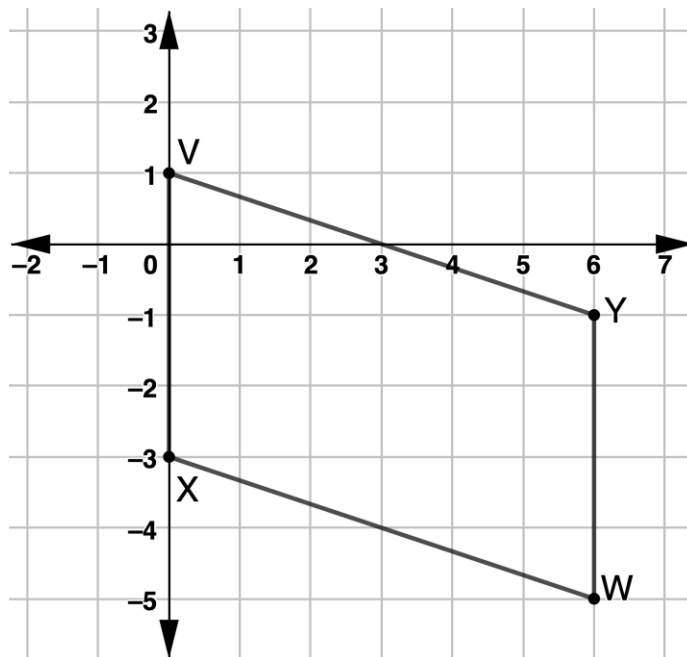
$$\frac{1}{\text{---}}V + \frac{1}{\text{---}}W$$

4. Find the point that partitions directed line segment VW in a 2:1 ratio.

Use the following information for questions 5-6. Line segment FW has endpoints at $F(-10, -10)$ and $W(6, -2)$.

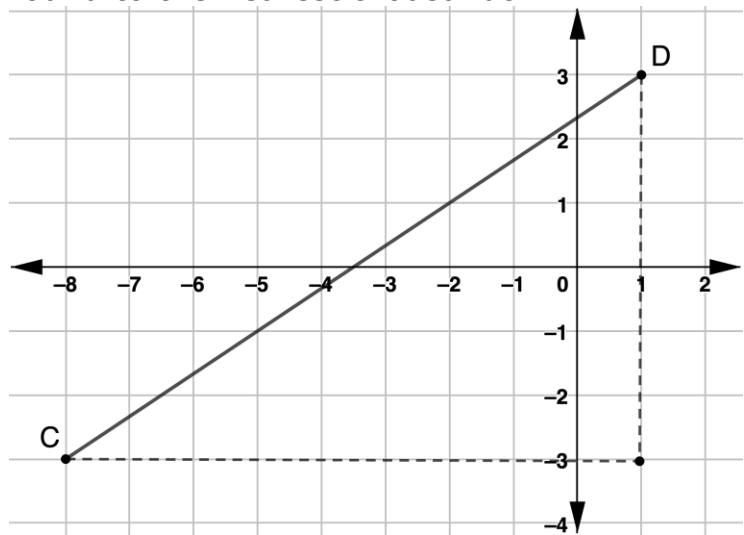
5. Write an expression for the point that partitions directed line segment FW in a ratio of 3 to 5.
6. Use the expression to find the point that partitions directed line segment FW in a 3:5 ratio.

Use the following information for questions 7-10. $VXWY$ has vertices at $V(0,1)$, $X(0,-3)$, $W(6,-5)$, and $Y(6,-1)$.



7. Find the point that partitions $\overline{WX}$ in a 1:4 ratio. Label it P .
8. Find the point that partitions $\overline{WY}$ in a 1:4 ratio. Label it F .
9. Find the point that partitions $\overline{XV}$ in a 1:4 ratio. Label it R .
10. Find the point that partitions $\overline{VY}$ in a 1:4 ratio. Label it T .

1. Find the length of segment CD using the Pythagorean Theorem. If necessary, round to the nearest thousandth.

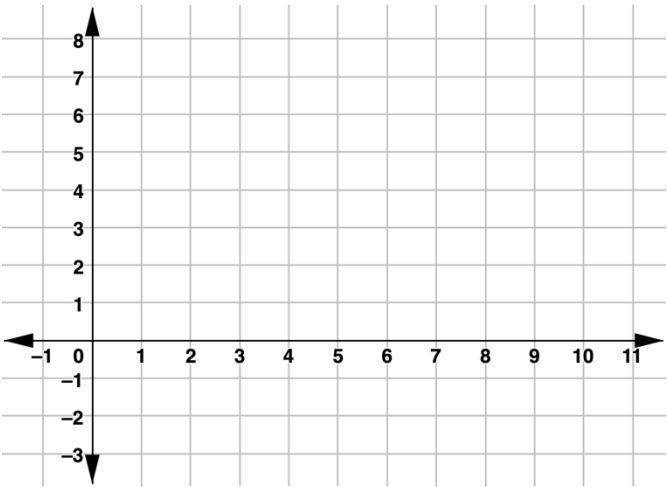


$CD =$ _____

Use the following information for questions 2-6.

$\triangle ABC$ has vertices at $A(9,6)$, $B(1,6)$, and $C(5,-3)$.

2. Graph $\triangle ABC$ on the coordinate plane.



Find the length of each segment. If necessary, round to the nearest tenth.

	Line Segment	Length of Line Segment
3.	$\overline{AB}$	
4.	$\overline{BC}$	
5.	$\overline{AC}$	

6. Complete the statement.

$\triangle ABC$ is classified as a(n)

- ☐ equilateral
- ☐ isosceles
- ☐ scalene

triangle because $\triangle ABC$ has

- ☐ no
- ☐ two
- ☐ three

equal side lengths.

Use the following information for questions 7-10.

$\triangle TUV$ has vertices at $T(1, -8)$, $U(8, -1)$, $V(4, -1)$.

Find the length of each segment. If necessary, round to the nearest tenth.

	Line Segment	Length of Line Segment
7.	$\overline{TU}$	
8.	$\overline{UV}$	
9.	$\overline{TV}$	

10. Complete the statement.

$\triangle TUV$ is classified as a(n)

- ☐ equilateral
- ☐ isosceles
- ☐ scalene

triangle because $\triangle TUV$ has

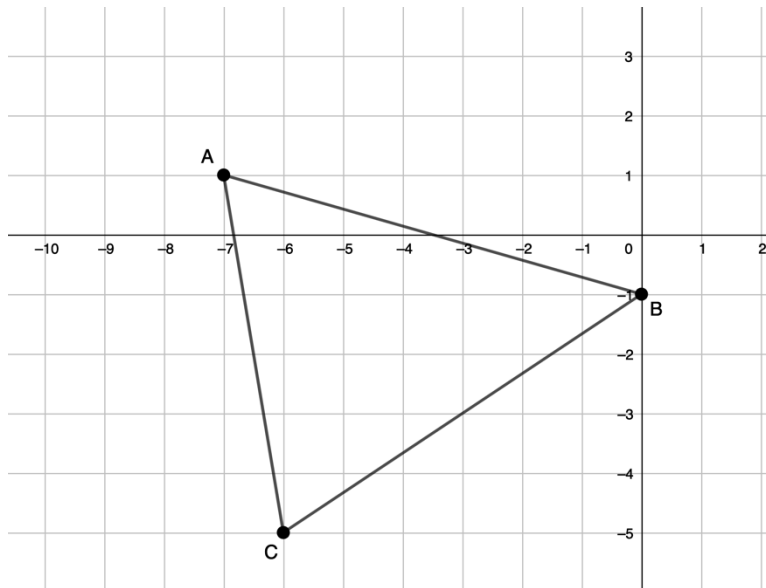
- ☐ no
- ☐ two
- ☐ three

equal side lengths.

Geometry
Unit 13
Lesson 7 Practice

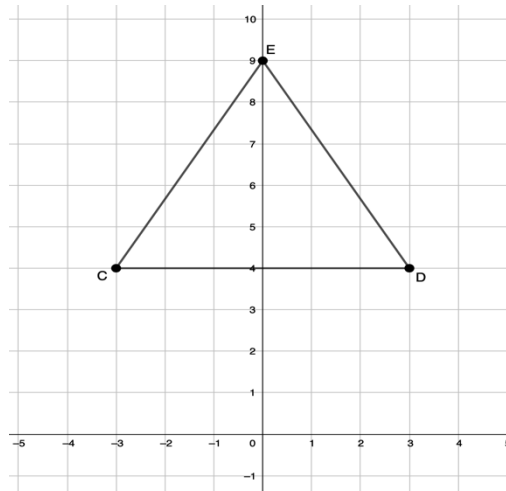
Name _____
Date _____
Period _____

Use the following information for questions 1-4. $\triangle ABC$ has vertices at $A(-7,1)$, $B(0,-1)$, and $C(-6,-5)$.



1. What is the average of the x -coordinates? Round to the nearest thousandth if necessary.
2. What is the average of the y -coordinates? Round to the nearest thousandth if necessary.
3. What are the coordinates of the centroid?
4. Verify the coordinates of the centroid by drawing the median of each side.

$\triangle CDE$ has vertices at $C(-3,4)$, $D(3,4)$, and $E(0,9)$.

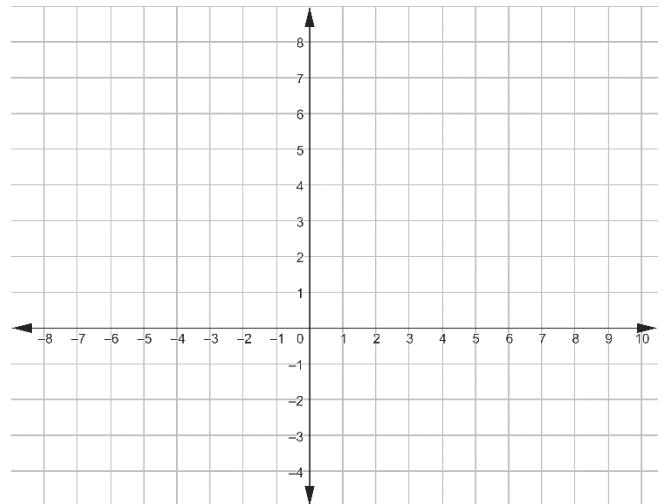


5. What is the average of the x -coordinates? Round to the nearest thousandth if necessary.
6. What is the average of the y -coordinates? Round to the nearest thousandth if necessary.
7. What are the coordinates of the centroid of $\triangle CDE$?
8. Verify the coordinates of the centroid by drawing the median of each side.

$\triangle NPQ$ has vertices at $N(8,3)$, $P(9,-3)$, and a centroid at $(\frac{10}{3}, \frac{4}{3})$.

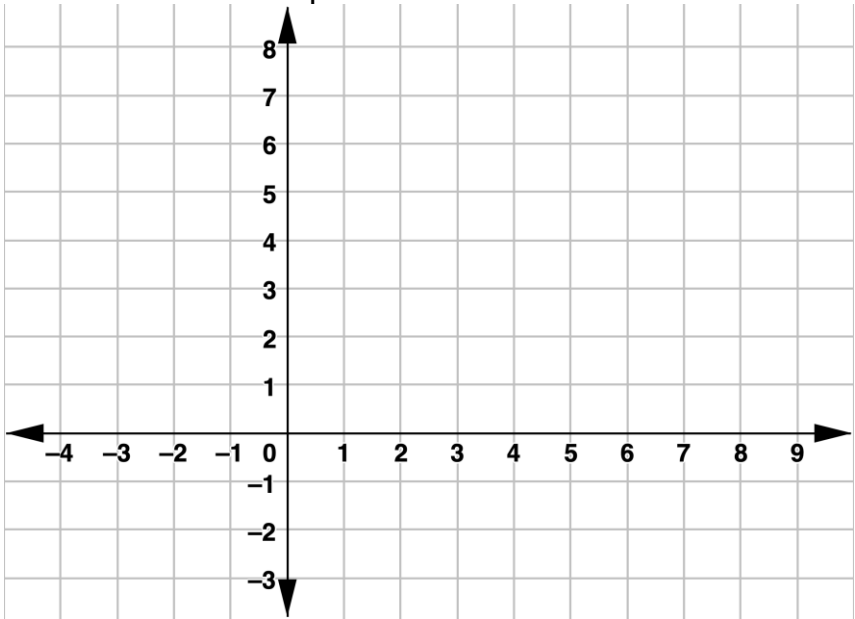
9. Graph $\triangle NPQ$ on the coordinate plane.

10. What are the coordinates of vertex Q ?



$MNSE$ has vertices at $M(6,7)$, $N(-4,3)$, $S(-2,-2)$, and $E(8,2)$.

1. Graph $MNSE$ on the coordinate plane.



2. Complete the table.

Line Segment	Length	Slope	Length of Diagonal Line Segment
$\overline{MN}$			
$\overline{NS}$			
$\overline{SE}$			
$\overline{ME}$			
$\overline{MS}$			
$\overline{NE}$			

3. Circle all classifications that apply to $MNSE$.

- Isosceles Trapezoid
- Kite
- Parallelogram
- Rectangle
- Rhombus
- Square
- Trapezoid

Complete the statements for questions 4 and 5.

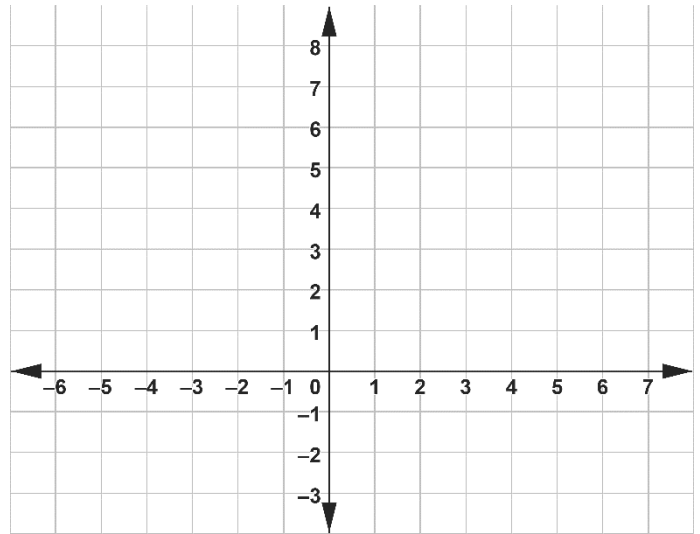
Since $MNSE$ 4.

☐ has
☐ does not have

 congruent diagonals, $MNSE$ is classified as a(n)

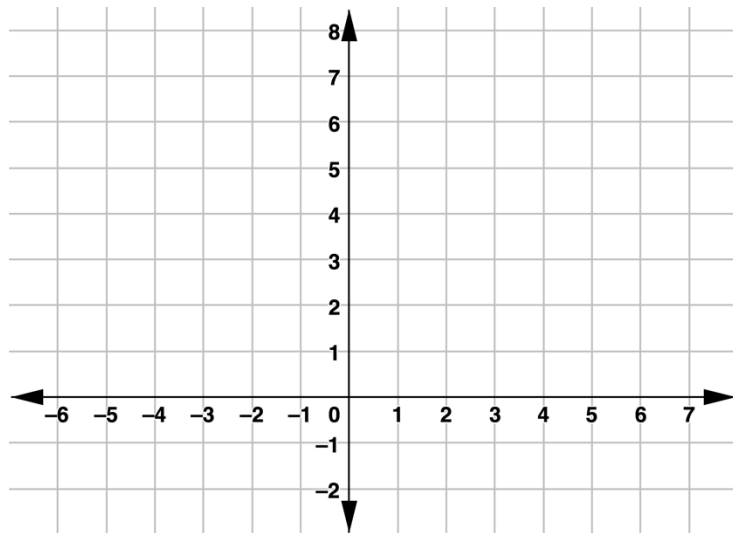
5. _____.

$YRDS$ has vertices at $Y(-6,5)$ $R(-2,-2)$,
 $D(6,-3)$, and $S(2,4)$.



6. $YRDS$ is classified as a _____ because...

Jeferson graphed $E(-2,8)$, $F(-6,4)$, $G(-4,2)$, $H(0,6)$ on the coordinate plane and classified the quadrilateral as a square.



7. Plot the coordinates Jeferson graphed for his quadrilateral.
8. The quadrilateral Jeferson graphed is a _____ because ...
9. Do you agree or disagree with Jeferson? If you disagree, explain a reason Jeferson could have used to classify the quadrilateral as a square.

Geometry
Unit 13
Lesson 9 Practice

Name _____
Date _____
Period _____

Najira is using a computer software to design the layout of her bathroom remodel. The base of the bathroom creates $DEFG$ with coordinates $D(7, -2)$, $E(1, -6)$, $F(3, -9)$, and $G(9, -5)$.

1. Complete the table. Show your work in the table. Leave answers in exact form.

Line Segment	Length
$\overline{DE}$	
$\overline{EF}$	
$\overline{FG}$	
$\overline{DG}$	

2. Complete the table. Show your work in the table. Leave answers in exact form.

Diagonal Line Segment	Length
$\overline{DF}$	
$\overline{EG}$	

3. Finish the statement.

$DEFG$ is classified as a _____ because ...

4. If 1 unit on the coordinate plane equals 3 feet (ft), what are the dimensions of the base of the bathroom? Round to the nearest thousandth.

5. $PRNT$ has vertices at $P(3, -8)$, $R(2, 0)$, $N(-5, 4)$, and $T(-4, -4)$.

$PRNT$ is classified as a(n) _____ because ...

Ginny is designing a template for her new 3D printer. She created two quadrilaterals using her software.

<i>MSFC</i>	<i>TDPR</i>
$M(0, -7), S(4, -3), F(6, 5), C(2, 1)$	$T(5, 7), D(7, 4), P(0, 4), R(2, 7)$

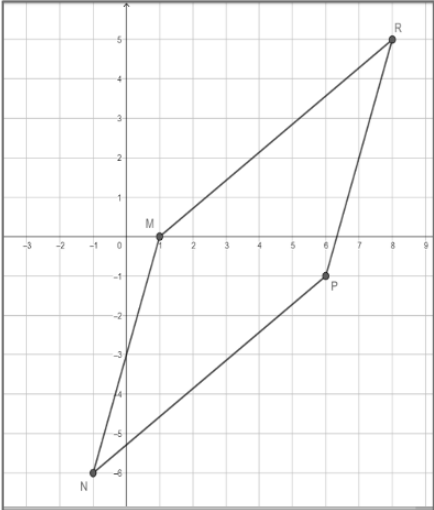
6. Which quadrilateral can be classified as a trapezoid? Show your work.
7. List the properties of a trapezoid that you used.
8. If 1 unit of the coordinate plane equals 4 inches (in.), what are the measurements of the sides of the trapezoid?

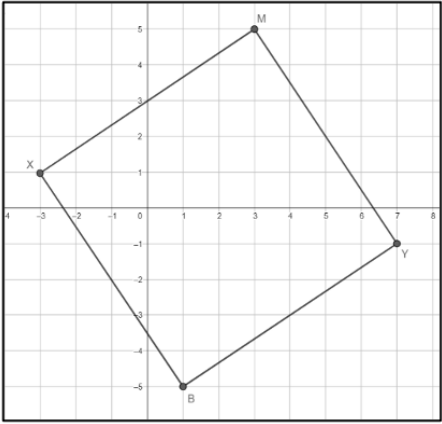
Fintan was graphing square $RD\overline{CP}$, but mixed up his vertices. He knows that point R has coordinates $(4, -5)$ and point P has coordinates $(2, -2)$. His two options for points C and D are shown.

Point C	Point D
$(-1, -4)$	$(-1, -7)$
$(-1, -3)$	$(1, -7)$

9. Which coordinates are correct for points C and D ? Justify your answer or show your work.
10. Which properties of a square did you use to determine the correct coordinates for points C and D ?

Complete the table for each image in questions 1 and 2.

Question 1	Work		Answer
			Properties
	Slope		
	Side Lengths		
	Diagonal Lengths		Quadrilateral

Question 2	Work		Answer
			Properties
	Slope		
	Side Lengths		
	Diagonal Lengths		Quadrilateral

For questions 3 and 4, given the coordinates, classify the quadrilaterals based on their properties.

Question 3	Work		Answer
			Properties
$CMHG$ has vertices at $C(9, -1)$, $M(2, 5)$, $H(-7, 7)$, and $G(0, 1)$. $CMHG$ is classified as a(n) _____ because.....	Slope		
	Side Lengths		
	Diagonal Lengths		Quadrilateral

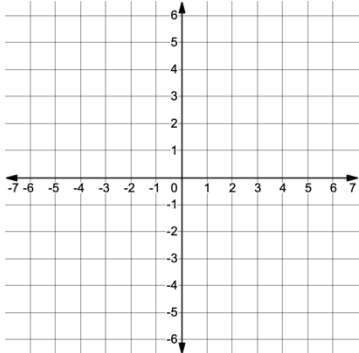
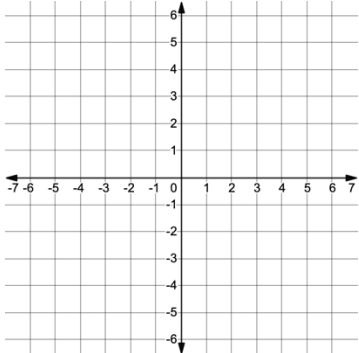
Question 4	Work		Answer
			Properties
$TCBY$ has vertices at $T(-3, -7)$, $C(5, -3)$, $B(3, 1)$, and $Y(-5, -3)$. $TCBY$ is classified as a(n) _____ because.....	Slope		
	Side Lengths		
	Diagonal Lengths		Quadrilateral

Geometry
Unit 13
Lesson 11 Practice

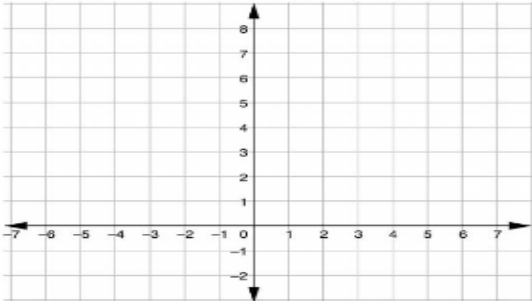
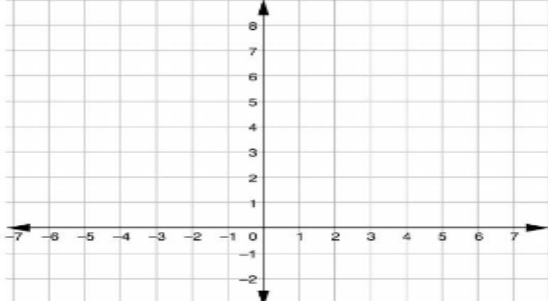
Name _____
Date _____
Period _____

Compare and contrast each image pair. Record your work and show your justifications in the table.

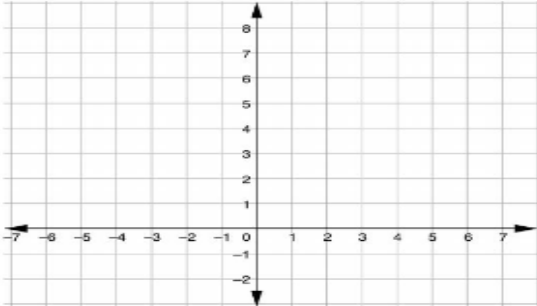
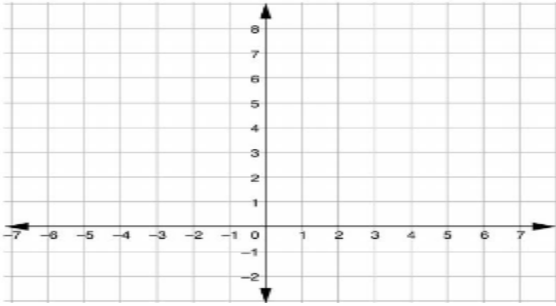
1.

Image 1	Image 2
	
Coordinates: $B(2,0)$, $C(-1,-5)$, $D(-6,-2)$	Coordinates: $B(3,1)$, $C(6,0)$, $D(4,-6)$
Classification:	Classification:
Property:	Property:
Similarity/Differences with Image 2:	Similarity/Differences with Image 1:

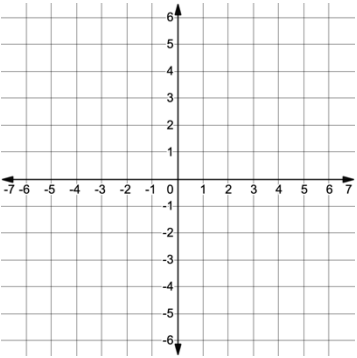
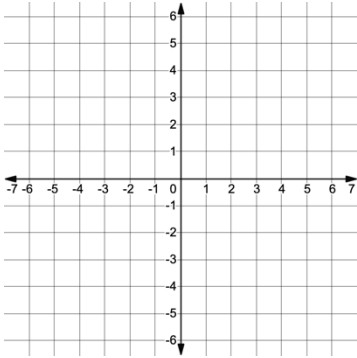
2.

Image 1	Image 2
	
Coordinates: $R(2,3)$, $S(6,7)$, $T(4,9)$, $V(0,5)$	Coordinates: $R(-6,5)$, $S(3,-1)$, $T(5,2)$, $V(-4,8)$
Classification:	Classification:
Property:	Property:
Similarity/Differences with Image 2:	Similarity/Differences with Image 1:

3.

Image 1	Image 2
	
Coordinates: $E(2, -1.5)$, $F(4, 2)$, $G(0.5, 4)$, $H(-1.5, 0.5)$	Coordinates: $E(7, 2)$, $F(3, 5)$, $G(-5, 7)$, $H(-1, 4)$
Classification:	Classification:
Property:	Property:
Similarity/Differences with Image 2:	Similarity/Differences with Image 1:

4.

Image 1	Image 2
	
Coordinates: $S(-6, -1)$, $T(-2, -3)$, $U(2, 0)$, $W(-3, 2)$	Coordinates: $S(5, 2)$, $T(5, 6)$, $U(6, 4)$, $W(1, 4)$
Classification:	Classification:
Property:	Property:
Similarity/Differences with Image 2:	Similarity/Differences with Image 1:

Right triangle MTH has vertices at $M(3, -8)$, $T(8, 4)$, $H(8, -8)$. Right triangle DPT has vertices at $D(-3, -4)$, $P(-5, -6)$, $T(-3, -6)$.

1. Determine if $\triangle MTH$ and $\triangle DPT$ are congruent, similar, or neither.

2. What definition, theorem, or property justifies your conclusions?

3. Describe the transformations that would show that the figures are congruent or that they are similar. State which image you used as the pre-image.

Determine if the given polygons are congruent, similar, or neither.

4. Trapezoid $TRPS$: $T(1, 1)$, $R(4, 5)$, $P(7, 5)$, $S(10, 1)$
Trapezoid $RTLF$: $R(3, 3)$, $T(12, 15)$, $L(21, 15)$, $F(30, 3)$
 - ☐ Congruent
 - ☐ Similar
 - ☐ Neither

5. Kite $GHKM$: $G(0, 0)$, $H(1, -4)$, $K(3, -3)$, $M(4, -1)$.
Kite $RSTV$: $R(0, 0)$, $S(2, -8)$, $T(6, -6)$, $V(8, -2)$.
 - ☐ Congruent
 - ☐ Similar
 - ☐ Neither

6. Rhombus $DEFG$: $D(6, 4)$, $E(-1, 0)$, $F(-2, -8)$, $G(5, -4)$
Rhombus $BKMH$: $B(3, 2)$, $K(-0.5, 0)$, $M(-1, -4)$, $H(2.5, -2)$
 - ☐ Congruent
 - ☐ Similar
 - ☐ Neither

Lane and Tonymy are determining if parallelograms $HMTY$ and $NRPX$ are congruent, similar, or neither. Parallelogram $HMTY$ has vertices at $H(5,6)$, $M(-3,3)$, $T(-4,-1)$, and $Y(4,2)$. Parallelogram $NRPX$ has vertices at $N(10,12)$, $R(-6,6)$, $P(-8,-2)$, and $X(8,4)$. Their work is shown.

Lane	Tonymy
Parallelograms $HMTY$ and $NRPX$ are similar because $\frac{MH}{RN} = \frac{XR}{PN}$	Parallelograms $HMTY$ and $NRPX$ are not similar because $\frac{MH}{RN} = \frac{XN}{PR}$

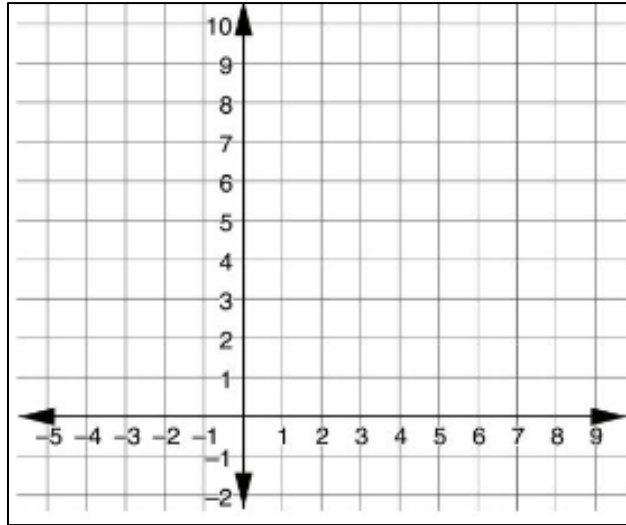
7. Who do you agree with?

8. Explain the mistake that was made and how to correct it.

Geometry
Unit 13
Lesson 13 Practice

Name _____
Date _____
Period _____

Parallelogram $FRGS$ has vertices at $F(-4,7)$, $R(-1,0)$, $G(6,-2)$, and $S(3,5)$.

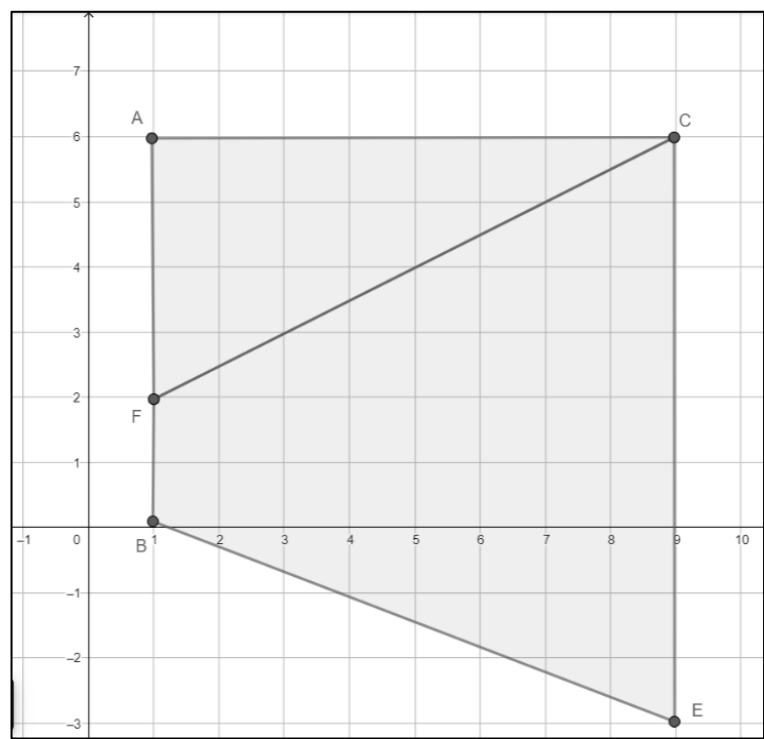


1. Graph $FRGS$.
2. Complete the table.

Line Segment	Length
$\overline{FR}$	
$\overline{RG}$	
$\overline{GS}$	
$\overline{FS}$	

3. What is the perimeter of $FRGS$?
4. What is the area of $FRGS$?

The composite figure shown is created by $A(1,6)$, $B(1,0)$, $C(9,6)$, $F(1,2)$, and $E(9,-3)$.



5. Complete the table. Round your answer to the nearest thousandth.

Figure	Perimeter	Area
$\triangle ACF$		
$BECF$		

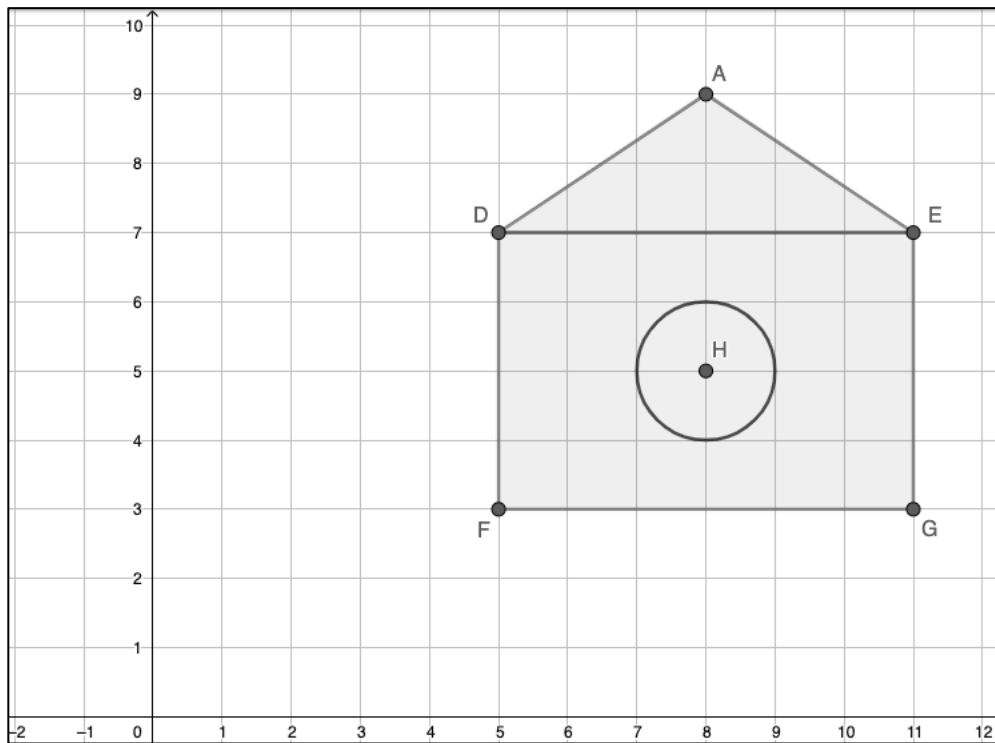
6. What is the perimeter of the composite figure?

7. What is the area of the composite figure?

Geometry
Unit 13
Lesson 14 Practice

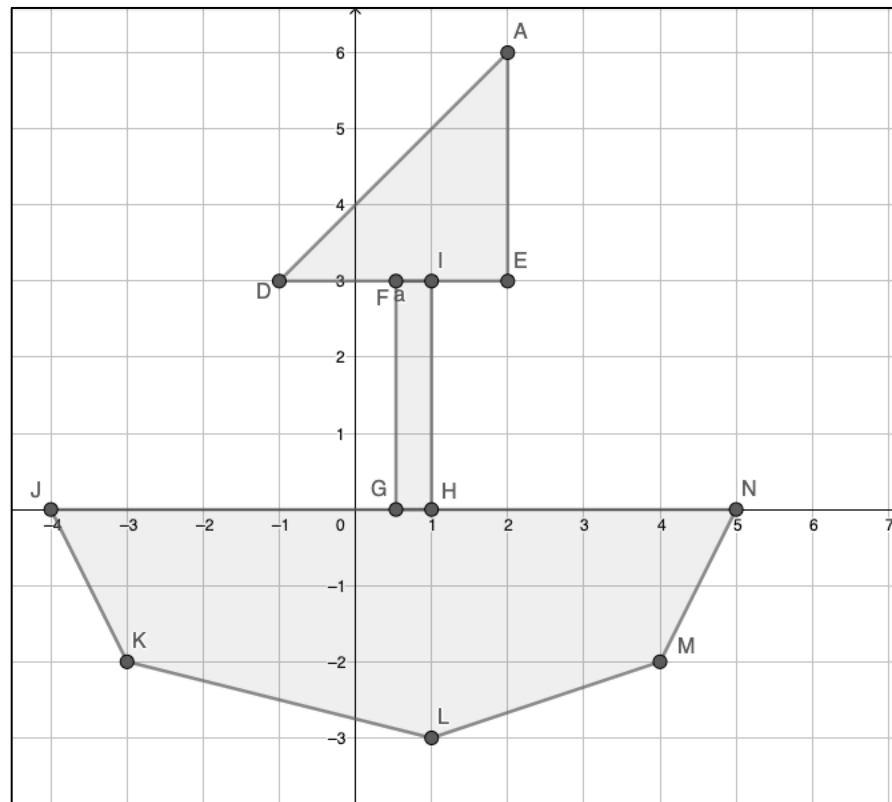
Name _____
Date _____
Period _____

Rory is designing the front of a birdhouse. The computer software generated coordinate points where each unit equals an inch. The roof has vertices at $A(8,9)$, $D(5,7)$, and $E(11,7)$. The front entrance has a circle with a radius of 1 unit. The front wall has vertices at $D(5,7)$, $E(11,7)$, $F(5,3)$, and $G(11,3)$.



1. Classify the type of triangle that is on the roof?
2. What is the total area of the birdhouse?
3. Cartwright wants to use weather-stripping around the entrance. It costs \$0.15 per inch. How much weather-stripping is going to be needed around the entrance to the bird house?

Kima and Whuan are building a sailboat by hand. The layout of the sailboat is sketched on the blueprint shown.



4. If one square on the grid represents two feet, what is the distance around the sailboard on the blueprint?

5. If one square on the grid represents three square feet, what is the total area of the sailboat on the blueprint?